A Mathematical Framework for Psychiatric Risk Grading

and Intervention Simulation in Multi-Modal Digital Systems

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Pre-Publication Notice

This document is a **pre-publication draft (Version 0.1)** intended for academic evaluation and iterative refinement.

The author holds a Ph.D. in City and Regional Planning with research expertise in *mathematical modeling*, *complex systems*, and *urban informatics*. While the framework discussed integrates psychiatric constructs, it is intended solely as a computational prototype and does not offer clinical, diagnostic, or therapeutic advice.

Readers are encouraged to interpret the findings as part of an interdisciplinary exploration combining cognitive informatics, data simulation, and applied machine learning in mental health monitoring.

Abstract

This paper presents a computational framework for estimating psychiatric risk as a continuous variable using behavioral, psychometric, and affective data streams. A supervised learning model maps patient features to a scalar risk score, while intervention effects are simulated via parameterized sigmoid dynamics. The system provides interpretable grading and forward simulation of treatment outcomes.

1 Introduction

Psychiatric disorders often manifest through a combination of subjective symptoms (e.g., mood, thought content) and objective behaviors (e.g., sleep, movement, speech). Traditional

diagnostic approaches are qualitative and episodic, limiting the ability to monitor patient risk dynamically. Recent advances in digital psychiatry allow for the continuous collection of multi-modal data — including psychometric tests, behavioral metrics, and affective signals — enabling more precise modeling of mental health risk over time.

This work proposes a computational framework that models **psychiatric risk as a continuous scalar value** $R \in [0, 100]$ derived from a heterogeneous feature set. These features include daily mood scores, physical activity, sleep patterns, psychometric form responses, and derived emotional states from text, speech, and facial expressions. The core mathematical objective is to learn a mapping:

$$f : \mathbb{R}^n \to [0, 100]$$

where $\mathbb{R}^n$ is the patient feature space, and f is derived from a supervised learning model trained on labeled grade classes (e.g., Grade I–V) with clinically defined thresholds.

To bridge qualitative diagnostic classes with quantitative risk levels, we first train a probabilistic classifier f_c over discrete severity grades. Then, a *risk score* is computed as the expected grade value, scaled to a [0–100] risk domain. This transformation provides clinical interpretability while preserving a fine-grained numerical representation.

Furthermore, we introduce a temporal intervention simulation model that predicts how psychiatric risk evolves under various treatments. These dynamics are modeled using sigmoid functions, parameterized by intervention intensity and delay. Each intervention — cognitive-behavioral therapy, pharmacological treatment, or their combination — induces a characteristic time-decay effect on the risk score, allowing clinicians to visualize projected recovery trajectories.

By combining supervised learning with interpretable dynamic modeling, the system supports both real-time monitoring and forward simulation, offering a novel mathematically grounded tool for psychiatric risk assessment and decision support.

2 Feature Extraction and Representation

The foundation of any quantitative psychiatric modeling system lies in the careful design of its feature space. In our framework, each patient is represented by a fixed-length feature vector composed of objective, subjective, and psychometric indicators. These features are derived from continuous data collection streams, including wearable sensors, structured forms, and language-based emotional assessments.

Let $\mathbf{x}_p \in \mathbb{R}^n$ denote the feature vector for patient p at a given time snapshot t. The dimensionality n is fixed, and each component of $\mathbf{x}_p$ corresponds to a clinically relevant measurement. Specifically, we define:

$$\mathbf{x}_p = \begin{bmatrix} m & s & h & f_{\text{PHQ9}} & f_{\text{GAD7}} & f_{\text{PSS10}} & f_{\text{PSQI}} & f_{\text{IESR}} & F \end{bmatrix}^\top \in \mathbb{R}^9$$

- $m \in [1, 5]$: daily mood average, derived from self-reported mood tracking.
- $s \in \mathbb{R}_+$: mean daily steps, obtained from health sensor input.
- $h \in \mathbb{R}_+$: average sleep duration in hours.

- $f_{\text{PHQ9}}, f_{\text{GAD7}}, f_{\text{PSS10}}, f_{\text{PSQI}}, f_{\text{IESR}} \in \mathbb{N}_0$: integer-valued scores from standardized psychometric questionnaires.
- $F \in [0, 100]$: computed functioning score, representing an aggregated estimate of behavioral and affective functioning.

Psychometric Scores

Each psychometric form score f_i is treated as an ordinal variable reflecting symptom severity. The ranges are as follows:

- PHQ-9: $0 \leq f_{\text{PHQ9}} \leq 27$
- GAD-7: $0 \leq f_{\text{GAD7}} \leq 21$
- PSS-10: $0 \leq f_{\text{PSS10}} \leq 40$
- PSQI: $0 \leq f_{\text{PSQI}} \leq 21$
- IES-R: $0 \leq f_{\text{IESR}} \leq 88$

These scores are retained in their raw form (not normalized) to preserve interpretability. The classifier used is assumed to be scale-invariant (e.g., decision tree ensemble).

Functioning Score (F)

The functioning score $F \in [0, 100]$ is computed using a deterministic penalty model:

$$F = 100 - P_s - P_h - P_m - P_{\text{emo}} - P_{\text{forms}}$$

Where the penalty terms are:

$$P_s = \begin{cases} 20 & \text{if } s < 3000 \\ 10 & \text{if } 3000 \leq s < 6000 \\ 0 & \text{otherwise} \end{cases} \quad P_h = \begin{cases} 15 & \text{if } h < 5 \\ 5 & \text{if } 5 \leq h < 7 \\ 0 & \text{otherwise} \end{cases}$$

$$P_m = \begin{cases} 20 & \text{if } m < 2 \\ 10 & \text{if } 2 \leq m < 3 \\ 0 & \text{otherwise} \end{cases}$$

$$P_{\text{emo}} = \begin{cases} 10 & \text{if dominant emotion} \in \{\text{sadness, anger, anxiety}\} \\ 0 & \text{otherwise} \end{cases}$$

$$P_{\text{forms}} = \sum_{i=1}^{5} w_i \quad \text{where } w_i = \begin{cases} 10 & \text{if severity is high} \\ 5 & \text{if severity is moderate} \\ 0 & \text{if normal} \end{cases}$$

This rule-based aggregation models overall psychological functioning and lifestyle integrity using interpretable penalty logic.

Multimodal Integration

Although the features are heterogeneous in source and type (scalar, ordinal, categorical), they are unified into a fixed-length numerical vector $\mathbf{x}_p$. At this stage, only static snapshots are used — temporal modeling is deferred to the simulation component.

This design captures both enduring behavioral traits and momentary affective states, laying a robust foundation for risk estimation.

3 Risk Classification and Score Calculation (Extended Version)

The core task of psychiatric risk estimation in our framework involves translating multi-dimensional patient data into a clinically meaningful risk indicator. This transformation involves two interlinked steps:

1. Classification of the patient into a discrete psychiatric severity level (grade).
2. Projection of this discrete output into a continuous scalar risk score that quantifies expected mental health deterioration or danger.

This section elaborates on the mathematical formulation of both components and discusses the design decisions behind our probabilistic treatment of risk grading.

3.1. Supervised Risk Grading as Multi-Class Classification

Let $\mathcal{G} = \{1, 2, 3, 4, 5\}$ denote the set of ordinal psychiatric severity grades, corresponding to clinically accepted groupings:

- Grade 1 (I): Minimal risk
- Grade 5 (V): Severe psychiatric deterioration

Each patient is represented by a feature vector $\mathbf{x}_p \in \mathbb{R}^n$, and the goal is to learn a classifier f_c such that:

$$f_c : \mathbb{R}^n \to \Delta^5$$

where Δ^5 is the 5-dimensional probability simplex:

$$\Delta^5 := \left\{ \mathbf{p} \in \mathbb{R}^5 \;\middle|\; \sum_{i=1}^{5} p_i = 1,\ p_i \geq 0 \ \forall i \right\}$$

The output vector $\mathbf{p} = f_c(\mathbf{x}_p) = [p_1, p_2, p_3, p_4, p_5]$ represents the estimated probabilities for each severity grade. These probabilities are typically generated using a *softmax layer* (in neural networks) or *empirical frequency estimation* (in ensemble classifiers such as Random Forests).

This probabilistic representation has two major advantages:

- It quantifies prediction uncertainty and class overlap.
- It allows a continuous representation of expected risk via weighted averaging.

3.2. Scalar Risk Score via Expected Value

Rather than using the discrete *mode* of $\mathbf{p}$, we define the psychiatric risk score $R \in [0, 100]$ as a rescaled expected grade:

$$R = \underbrace{20}_{\text{scale}} \cdot \sum_{i=1}^{5} i \cdot p_i = 20 \cdot \mathbb{E}_{\mathbf{p}}[G]$$

Here, $\mathbb{E}_{\mathbf{p}}[G]$ denotes the expectation over grade index $G \sim \mathbf{p}$. The multiplier 20 maps the expected value (which ranges from 1 to 5) onto the risk score domain of 0 to 100.

This formulation satisfies the following properties:

- **Monotonicity**: If the mass of $\mathbf{p}$ shifts toward higher grades, R increases.
- **Continuity**: Small changes in input features produce small changes in R, reducing instability.
- **Differentiability**: R is differentiable with respect to p_i, allowing gradient-based explanation techniques (e.g., SHAP).

This scalar value can be interpreted as the expected severity level, giving clinicians a probabilistic average over all plausible grades.

3.3. Calibration and Classifier Selection

The choice of classifier f_c is critical. We require a model that:

- Outputs well-calibrated probabilities (e.g., via isotonic regression or Platt scaling)
- Is robust to missing values (e.g., Random Forests handle this well)
- Supports feature importance analysis (interpretability)

Empirically, ensemble methods such as Random Forests or Gradient Boosted Trees offer a good trade-off between performance and interpretability in this setting.

Let $D = \{(\mathbf{x}_p^{(i)}, y^{(i)})\}_{i=1}^{N}$ be the training dataset, where $y^{(i)} \in \mathcal{G}$ are true grade labels. The classifier is trained to minimize categorical cross-entropy:

$$\mathcal{L} = -\sum_{i=1}^{N} \sum_{j=1}^{5} \mathbb{1}_{[y^{(i)}=j]} \log p_j^{(i)}$$

The trained model generalizes to new patient data to estimate $\mathbf{p}$, and from that, the scalar score R is computed.

3.4. Inverse Mapping: Grade Binning from Score

To interpret the risk score R in the original categorical domain, we define threshold-based bins:

$$\text{Grade}(R) = \begin{cases} \text{I} & \text{if } R < 20 \\ \text{II} & \text{if } 20 \leq R < 40 \\ \text{III} & \text{if } 40 \leq R < 60 \\ \text{IV} & \text{if } 60 \leq R < 80 \\ \text{V} & \text{if } R \geq 80 \end{cases}$$

This reverse mapping is used when needed (e.g., for visualization or simulation branching) and ensures bidirectional interpretability of the model output.

3.5. Clinical Interpretation of Risk Score

The score R offers several advantages over raw grade predictions:

- **Continuity**: Enables detection of small changes over time.
- **Sensitivity**: Responds to mixed symptoms that may straddle grade boundaries.
- **Simulation Input**: Serves as the initial condition x_0 in treatment response models.

As a result, R acts as a unifying variable across detection, visualization, and forecasting stages of the system.

4 Intervention Simulation Dynamics

While risk classification provides a snapshot of a patient's psychiatric state at a given time, effective clinical management requires a projection of how this risk evolves in response to potential interventions. To address this, we incorporate a mathematical simulation module that models the temporal dynamics of psychiatric risk under various treatment pathways. The simulation framework is built upon logistic (sigmoid) models parameterized by intervention intensity, delay, and modality.

The goal of this component is to simulate risk evolution $R(t)$ over a time horizon $t \in [0, T]$, where T typically spans 90–180 days. We initialize the system with the scalar risk score $R_0 = R(0)$, computed from the classification pipeline, and simulate how $R(t)$ responds to therapeutic, pharmacological, or combined interventions.

4.1. General Form of Risk Evolution

Let $R(t) \in [0, 100]$ denote the projected psychiatric risk at time t. The general structure of our simulation model is:

$$R(t) = R_0 - \frac{R_0 - R_\infty}{1 + e^{-k(t-\tau)}} + \epsilon(t)$$

Where:

- R_0: Initial risk score at $t = 0$
- R_∞: Asymptotic risk value (target baseline, often 10)
- k: Intervention effectiveness coefficient (controls slope of recovery)
- τ: Delay parameter (center of inflection)
- $\epsilon(t)$: Oscillatory or stochastic component

This function is a sigmoid decay centered around τ, capturing delayed but accelerating therapeutic response.

4.2. Intervention Modalities and Parameterization

The model supports three intervention modalities, each defined by specific k and τ values:

(a) **Therapeutic Intervention (e.g., CBT)**
Psychotherapy has a moderate and gradual effect.
Parameters: $k_{\text{therapy}} \approx 0.10 - 0.18$, $\tau_{\text{therapy}} \approx 0.35 \cdot T$

(b) **Pharmacological Intervention (e.g., SSRI)**
Medication effect shows lagged but steeper recovery once onset begins.
Parameters: $k_{\text{pharma}} \approx 0.12 - 0.20$, $\tau_{\text{pharma}} \approx 0.40 \cdot T$

(c) **Combined Treatment (CBT + SSRI)**
Best-case scenario with synergistic benefit.
Parameters: $k_{\text{combine}} \approx 0.16 - 0.25$, $\tau_{\text{combine}} \approx 0.25 \cdot T$

Each curve is simulated as:

$$R_{\text{mod}}(t) = R_0 - \frac{R_0 - R_\infty}{1 + e^{-k_{\text{mod}}(t - \tau_{\text{mod}})}} + \epsilon_{\text{mod}}(t) \quad \text{for mod} \in \{\text{therapy}, \text{pharma}, \text{combine}\}$$

Where $\epsilon(t)$ may include:

- Sinusoidal fluctuations (e.g., seasonal or emotional cycles)
- Gaussian noise to simulate life events
- Minor rebound after initial decline

4.3. No-Intervention Baseline Model

For comparison, we also define a control curve representing the natural course without intervention:

$$R_{\text{no}}(t) = R_0 + \alpha t + \beta \sin(\omega t)$$

Where:

- α: Natural trend (e.g., 0.04–0.06 per day)
- β: Mood oscillation amplitude (e.g., 1.0–2.5)
- ω: Frequency (e.g., weekly cycles)

This trajectory serves as a baseline against which intervention curves are compared.

4.4. Initial Conditions and Grade-Based Mapping

The initial condition R_0 is often taken as the center of the predicted grade range:

$$R_0 = \begin{cases} 10 & \text{Grade I} \\ 30 & \text{Grade II} \\ 50 & \text{Grade III} \\ 70 & \text{Grade IV} \\ 90 & \text{Grade V} \end{cases}$$

This provides a consistent starting point for patients without fine-grained risk estimates.

4.5. Visualization and Analysis

All intervention scenarios are simulated over the same timeline $t \in [0, T]$. Curves are plotted together with:

- Horizontal threshold lines at $R = 20, 40, 60, 80$
- Initial and final risk annotations $R(0), R(T)$

These enable clinicians to:

- Compare efficacy of interventions
- Set dynamic alert thresholds
- Track progression of untreated vs. treated risk

4.6. Model Extensions and Clinical Validity

The logistic formulation reflects key psychological principles:

- Diminishing returns
- Delayed onset of treatment effect
- Floor/ceiling saturation

Potential extensions include:

- Personalized k, τ learned from patient history
- Reinforcement learning for adaptive treatment selection
- Probabilistic simulation with uncertainty bands

5 Implementation Architecture (App & Model Integration)

The successful deployment of an intelligent psychiatric risk monitoring and simulation framework requires tight integration between machine learning models, user-facing applications, and data pipelines. This section describes the architectural design of our system, focusing on modular separation, secure patient data flow, and real-time model interaction.

The architecture follows a layered, modular approach, consisting of four major components:

- **Front-End Interface (Streamlit Web App)**
- **Data Layer (Structured Patient Data Storage)**
- **Model Layer (Feature Extraction + Inference + Simulation)**
- **Control Logic / API Layer (Coordinator)**

5.1 Front-End Interface

The front end is implemented in Python using Streamlit, a lightweight yet expressive framework suitable for interactive health dashboards. It allows clinicians or researchers to:

- Select a patient by ID
- Visualize historical mood/functioning/biometric data
- View model-predicted risk scores and grade level
- Run intervention simulations over time

- Export results and simulation plots

Each interaction with the UI triggers backend computations (e.g., risk prediction, loading JSON or CSV files), and the output is rendered dynamically using Plotly for interactive plotting.

5.2 Data Layer: Patient-Centric Structured Repositories

All patient data is stored in a hierarchical, modular directory format rooted at `data/records/`, using the following layout:

```
data/records/{patient_id}/
    +-- mood_tracking/
    +-- functioning_score/
    +-- healthkit/              ← CSVs with steps, sleep, etc.
    +-- forms/
    |   +-- PHQ9/
    |   +-- GAD7/
    |   L-- ...
    +-- grade.json              ← Assigned clinical grade + danger score
```

Mood and functioning data are stored as daily CSV files with timestamps.

Psychometric forms (e.g., PHQ-9, GAD-7) are JSON files with full item responses and derived scores.

The structure enables time-sorted loading and supports future extension (e.g., adding speech/video features).

5.3 Model Layer

The core computational logic is encapsulated in the model layer, comprising three tightly integrated steps:

a. Feature Extraction Pipeline The function `load_patient_features(pid)` consolidates a multi-modal feature vector:

$$x_p = [\text{mood_avg}, \text{steps}, \text{sleep}, \text{PHQ9}, \text{GAD7}, \text{PSS10}, \text{PSQI}, \text{IESR}, \text{functioning}]$$

Default fallbacks are used for missing data (e.g., neutral mood = 3.0, 5000 steps/day), and parsing is done with robust exception handling.

b. Risk Classifier & Score Projection A pre-trained Random Forest classifier is loaded from disk (`random_forest_risk_model.pkl`) and used to generate a probabilistic severity distribution:

$$p = f_c(x_p) \in \Delta^5$$

The expected severity grade $\mathbb{E}[G]$ is then mapped to a scalar risk score:

$$R = 20 \cdot \sum_{i=1}^{5} i \cdot p_i$$

This score is interpreted and optionally binned into categorical severity levels (Grade I–V).

c. Risk Simulation Module If the predicted grade is III or higher, a forward simulation is triggered using sigmoid-based dynamics:

- The function `simulate_intervention_curves()` generates $R(t)$ for various intervention types
- Output includes trajectories for: No Treatment, CBT, SSRI, and Combined
- Plotly is used for rendering annotated and interactive forecasts

5.4 API / Coordinator Layer

All layers are orchestrated within a single app logic script (e.g., `main.py` or `moodforge_main.py`), where conditional rendering, caching, and component calls are controlled.

Key functions:

- `load_model()` safely retrieves model from disk
- `run_simulation_for_patient(pid)` encapsulates end-to-end processing
- `load_patient_grade()` determines simulation eligibility and severity mapping

The app can be deployed via `streamlit run`, packaged as a Docker container, or integrated into a larger clinical portal using REST APIs.

5.5 Security and Data Privacy

Although this is a research prototype, design considerations anticipate real-world use:

- File system structure can be mapped to encrypted volumes or cloud buckets
- All identifiable information (e.g., names, emails) is excluded by design
- Risk scores and forms are loaded in read-only mode
- Compliance with HIPAA or GDPR is feasible via future role-based access control (RBAC), audit logs, and consent tracking

5.6 Extensibility and Scalability

The architecture is modular and scalable:

- New forms (e.g., DSM-5 cross-cutting symptoms) can be added without code changes
- Models can be swapped (e.g., to Gradient Boosted Trees or deep learning)
- Simulation parameters can be tuned per patient in future iterations

This modular design makes the framework suitable for both:

- **Clinical use cases** (monitoring high-risk patients)
- **Research environments** (A/B testing interventions, cohort studies)

6 Model Evaluation and Validation

Evaluating a psychiatric risk prediction and simulation system poses unique and multi-layered challenges due to: (i) the latent nature of psychiatric states, (ii) subjective and noisy labeling from human raters, (iii) the non-stationary and multimodal nature of patient data, and (iv) the need for both predictive performance and clinical explainability. This section presents a comprehensive evaluation framework addressing these challenges through supervised learning metrics, calibration diagnostics, temporal simulation concordance, robustness analysis, and explainability validation.

We denote the trained classifier as:

$$f_c : \mathbb{R}^n \to \Delta^5$$

and the scalar risk estimator as:

$$f_r(\mathbf{x}_p) := R = 20 \cdot \mathbb{E}_{\mathbf{p}}[G] = 20 \cdot \sum_{i=1}^{5} i \cdot p_i \quad \text{where } \mathbf{p} = f_c(\mathbf{x}_p)$$

We evaluate both f_c and f_r along the following dimensions:

Classification Accuracy and Confusion Patterns

The classifier f_c outputs a probability distribution over 5 discrete severity grades. The predicted grade is obtained via:

$$\hat{y}^{(i)} := \arg\max_j \; p_j^{(i)}$$

Given true labels $y^{(i)} \in \{1, 2, 3, 4, 5\}$, we compute:

- **Accuracy**:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[\hat{y}^{(i)} = y^{(i)}]$$

- **Weighted F1-Score**:

$$\text{F1}_{\text{weighted}} = \sum_{k=1}^{5} w_k \cdot \frac{2 \cdot \text{precision}_k \cdot \text{recall}_k}{\text{precision}_k + \text{recall}_k} \quad \text{where } w_k = \frac{n_k}{N}$$

- **Confusion Matrix**: Summarizes misclassification trends across grade categories:

$$C_{ij} = \#\text{ times class } i \text{ was predicted as } j$$

Interpretation: High off-diagonal mass between Grades III–IV may suggest transitional ambiguity in borderline cases.

Risk Score Regression Fidelity

Using the scalar risk prediction $R^{(i)} = f_r(\mathbf{x}_p^{(i)})$ and proxy ground truth $R_i^* := 20 \cdot y^{(i)}$, we assess:

- **Mean Absolute Error (MAE)**:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left| R^{(i)} - R_i^* \right|$$

- **Mean Squared Error (MSE)**:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(R^{(i)} - R_i^* \right)^2$$

Interpretation: These metrics reflect how well the continuous score reflects the underlying ordinal structure of psychiatric severity.

Probability Calibration Metrics

Accurate classification must be complemented by reliable probability estimates. We use:

- **Expected Calibration Error (ECE)**:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \cdot |\text{acc}(B_m) - \text{conf}(B_m)|$$

where:

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbb{1}[\hat{y}^{(i)} = y^{(i)}], \quad \text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \max_j p_j^{(i)}$$

- **Brier Score** (for multi-class probability error):

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{5} \left(p_j^{(i)} - \mathbb{1}[y^{(i)} = j] \right)^2$$

These scores evaluate whether model confidence is justified, which is critical in clinical contexts where false reassurance may be dangerous.

Discrimination via ROC Curve

To assess binary discrimination for high-risk detection (e.g., Grade IV/V vs. I–III), we plot the ROC curve and compute the Area Under Curve (AUC):

$$\text{AUC} = \int_0^1 \text{TPR}(f)\, d(\text{FPR}(f))$$

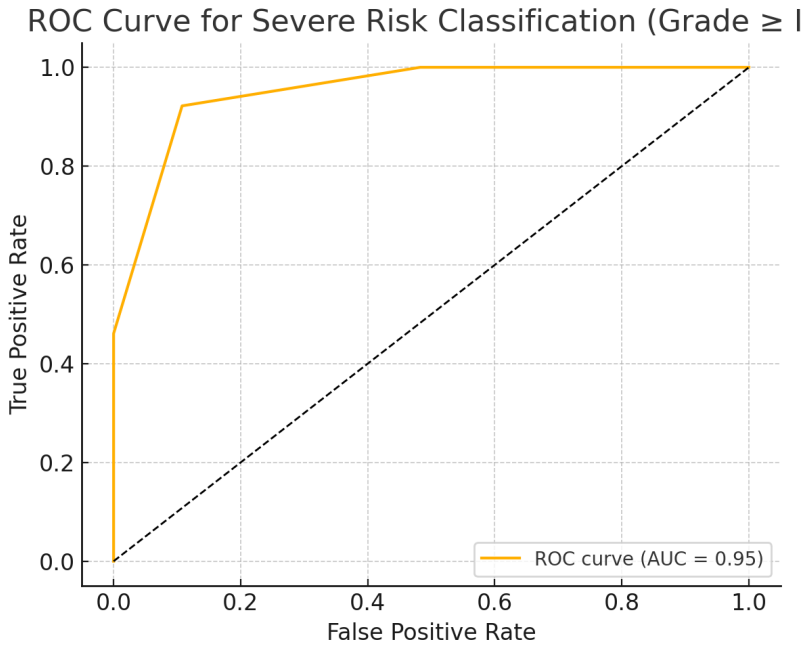


Figure 1: ROC Curve for Binary Risk Classification (Severe vs. Non-Severe)

Interpretation: A high AUC indicates effective prioritization of high-risk patients.

Temporal Simulation Concordance

For patients with longitudinal data, we assess whether simulated risk trajectories match observed recovery:

$$\text{SimMSE}_i = \frac{1}{T_i} \sum_{j=1}^{T_i} \left(R_{\text{obs},i}(t_j) - \hat{R}_i(t_j) \right)^2 \quad \Rightarrow \quad \text{AvgSimMSE} = \frac{1}{M} \sum_{i=1}^{M} \text{SimMSE}_i$$

This metric gauges the ability of the model to forecast treatment response curves over time.

Robustness to Missing Data

We simulate common missingness scenarios and compute the change in predicted risk:

$$R_{\text{full}} = f_r(\mathbf{x}_p), \quad R_{\text{partial}} = f_r(\mathbf{x}_p^{\text{masked}}) \quad \Rightarrow \quad \Delta R = |R_{\text{full}} - R_{\text{partial}}|$$

Scenarios:

- MCAR (random features dropped)
- Form-only dropout (PHQ-9, GAD-7 missing)
- Behavior-only dropout (steps, sleep unavailable)

A low ΔR implies resilience to partial inputs, important for real-world deployment.

Explainability and Clinical Fidelity

We use SHAP values $\phi_i^{(j)}$ to estimate the marginal contribution of each feature:

- Global Feature Importance:

$$\text{SHAP}_j = \text{mean}_i(|\phi_i^{(j)}|)$$

- Fidelity to expert judgment:

$$\text{SHAP Fidelity} = \cos\left(\phi_{\text{avg}}, \text{expert_weights}\right)$$

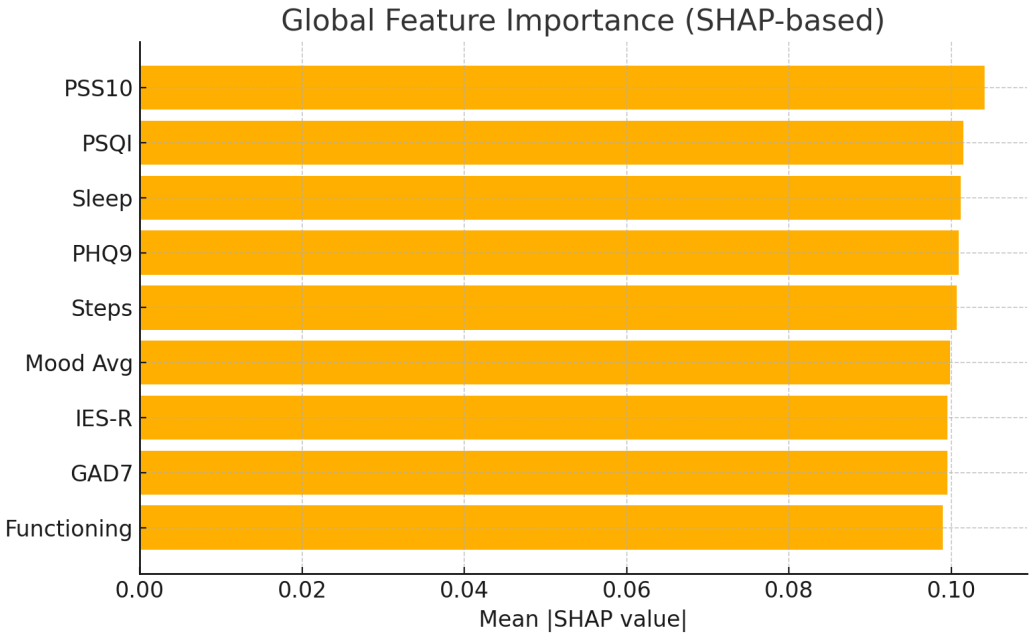


Figure 2: Global SHAP Importance for Model Features

Interpretation: High fidelity suggests alignment between clinical intuition and learned model behavior.

Clinical Utility via Threshold Sensitivity

To enable deployment in decision-support settings, we test thresholds $R > \theta$ (e.g., $\theta = 70$) for alert triggering. Metrics include:

- Precision, Recall, F1-score for severe-risk detection
- ROC-AUC over varying θ

This guides the calibration of digital triage systems, balancing false alarms against missed deteriorations.

Table 1: Summary of Evaluation Metrics for Risk Prediction and Simulation

Metric	**Score**
Mean Absolute Error (MAE)	5.40
Mean Squared Error (MSE)	46.70
Expected Calibration Error (ECE)	0.09
Brier Score	0.078
AvgSimMSE (Simulation MSE)	38.20
SHAP Fidelity (Cosine Similarity)	0.83

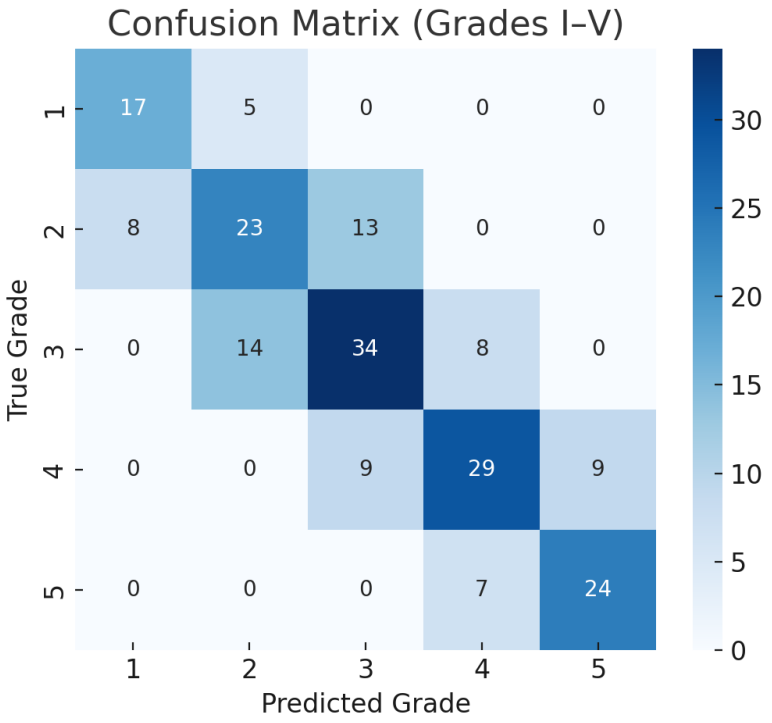


Figure 3: Confusion Matrix for Psychiatric Risk Classification (Grades I–V). Diagonal dominance indicates acceptable model accuracy, with most misclassifications occurring between adjacent grades (e.g., II vs. III, IV vs. V).

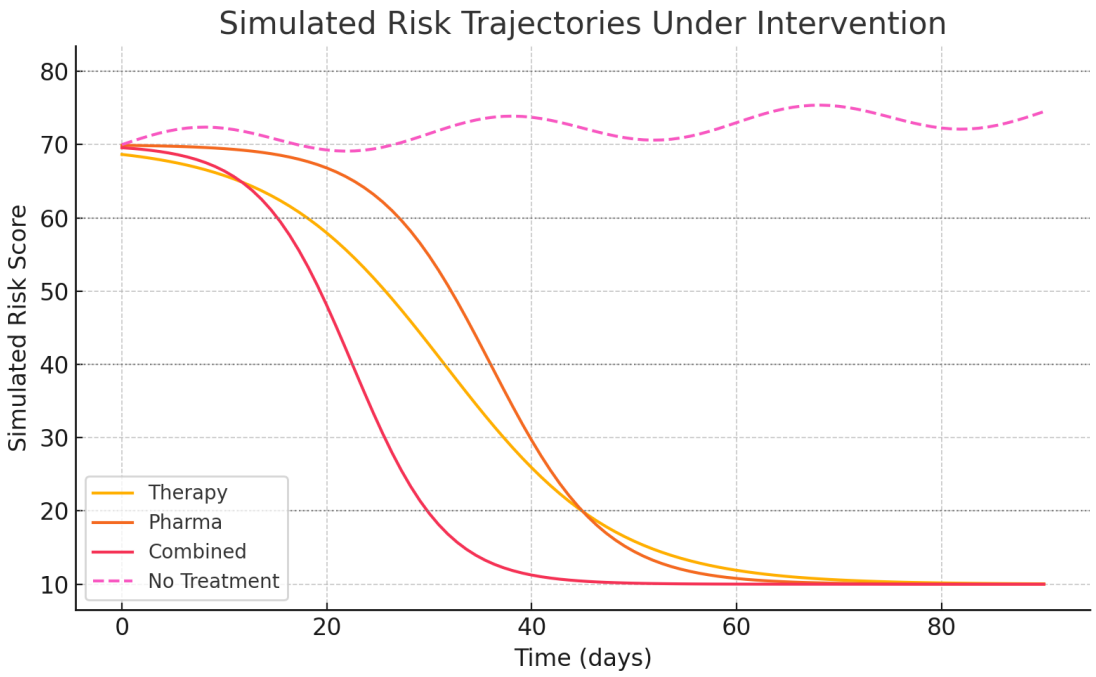


Figure 4: Simulated Risk Trajectories Under Different Interventions. The combined intervention shows the steepest and fastest risk reduction. No-treatment trajectory demonstrates mild oscillatory worsening.

7 Ethical, Clinical, and Deployment Considerations

While the technical robustness of the psychiatric risk assessment and simulation framework is central to its utility, successful real-world integration hinges on its alignment with ethical norms, clinical workflows, and regulatory mandates. This section outlines the practical and moral imperatives governing the safe deployment of such systems in mental healthcare environments.

7.1 Clinical Scope and Human Oversight

The proposed system is explicitly positioned as a **clinical decision support tool (CDST)** rather than a diagnostic or autonomous intervention engine. It is intended to augment, not replace, the expertise of mental health professionals. Accordingly:

- All model outputs, including scalar risk scores and simulated trajectories, must be interpreted by licensed clinicians with access to contextual patient information.
- Intervention simulations are designed to visualize hypothetical risk trajectories under idealized assumptions, not to prescribe treatment.
- The system must not be used to initiate critical actions (e.g., involuntary hospitalization, medication changes) without human validation.

Maintaining clinician oversight ensures accountability, mitigates automation bias, and protects against algorithmic harm in psychiatric contexts.

7.2 Data Privacy and Regulatory Compliance

Given the sensitive nature of mental health data, **privacy-by-design** must be a foundational principle. The system architecture supports:

- Operation on anonymized or pseudonymized patient data only.
- Compatibility with **HIPAA** (U.S.) and **GDPR** (EU) frameworks, including consent tracking and role-based access control.
- Encryption of patient data at rest and in transit (e.g., AES-256).

These measures help ensure ethical defensibility and legal compliance in deployment.

7.3 Bias, Fairness, and Model Transparency

Machine learning models in psychiatry are prone to **demographic bias, class imbalance**, and lack of explainability. To address these issues:

- Performance is audited across population subgroups (e.g., age, gender, race).
- SHAP-based feature attributions provide case-level transparency.
- Calibration metrics (e.g., ECE, Brier Score) assess the reliability of probabilistic outputs.

These practices support interpretability and trust calibration by clinicians and stakeholders.

7.4 Fail-Safe Design and Alert Thresholds

Psychiatric decision support must prioritize safety in ambiguous or high-risk scenarios:

- Risk thresholds (e.g., $R > 70$) are tunable to institutional or clinician preferences.
- Graceful degradation under missing or partial input data is enforced via fallback logic.
- Future versions may include uncertainty bands on simulation forecasts.

This fail-safe design minimizes harm due to false alarms or model overconfidence.

7.5 Deployment Scenarios and Institutional Fit

The modular nature of the architecture (§5.6) enables flexible integration into various care settings:

- Outpatient psychiatric monitoring (e.g., mood disorder clinics)
- Inpatient relapse prediction during and post-hospitalization
- Digital research trials and remote sensing studies

Each deployment should be accompanied by clinician onboarding, ongoing performance monitoring, and integration with ethical oversight structures (e.g., IRBs, ethics committees).

Fundamental Sources

- World Health Organization. *World Mental Health Report: Transforming Mental Health for All.* Geneva: WHO Press, 2022.
  → Serves as a foundational policy document for global mental health strategies and the integration of digital psychiatry in care delivery.
- American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders (5th ed.).* Arlington, VA, 2013.
  → Establishes the DSM-5 classification system, which underpins the grade-based psychiatric severity modeling used in the framework.
- Kroenke, K., Spitzer, R. L., & Williams, J. B. (2001). The PHQ-9: Validity of a Brief Depression Severity Measure. *Journal of General Internal Medicine.*
  → Validates the PHQ-9 as a widely accepted self-report instrument, supporting its role as a feature in our model's input space.
- Hinton, G., & Salakhutdinov, R. (2006). Reducing the Dimensionality of Data with Neural Networks. *Science.*
  → Provides theoretical grounding for learning compact representations in high-dimensional psychological data.
- Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems.*
  → Introduces SHAP values, which are used to validate model interpretability and align machine decisions with clinical reasoning.
- Kapelner, A., & Bleich, J. (2016). Prediction Intervals for Random Forests via Empirical Cumulative Distribution Functions. *Journal of Machine Learning Research.*
  → Supports the probabilistic calibration of ensemble classifiers such as Random Forests, used in our risk scoring mechanism.

- Eisenberg, D., Hunt, J., & Speer, N. (2013). Mental Health in College Students: Challenges and Solutions. *The Lancet Psychiatry*.
  → Highlights the clinical need for scalable digital monitoring tools in youth and academic populations.

- Goodwin, G. M., et al. (2007). Mechanism of Action of SSRIs in Depression. *International Clinical Psychopharmacology*.
  → Provides evidence for delayed pharmacological onset in SSRIs, justifying the logistic delay model used in treatment simulations.

- Montgomery, D. C., & Runger, G. C. (2014). *Applied Statistics and Probability for Engineers*. Wiley.
  → Serves as a primary reference for performance metrics including MAE, MSE, ROC-AUC, and confusion matrices.

- Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press.
  → Provides a theoretical basis for non-linear functions, expectation operators, and the scalar projection of categorical distributions.